Estimating Nonlinear Insulin-Glucose Dynamics in Critically-ill Patients

Markus Mulvihill

KTH Royal Institute of Technology

September, 2026

Thesis Presentation

2026-09-08

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2. Examiner - Ragnar Thobaben
3. Opponent - Hugo Sacilotto

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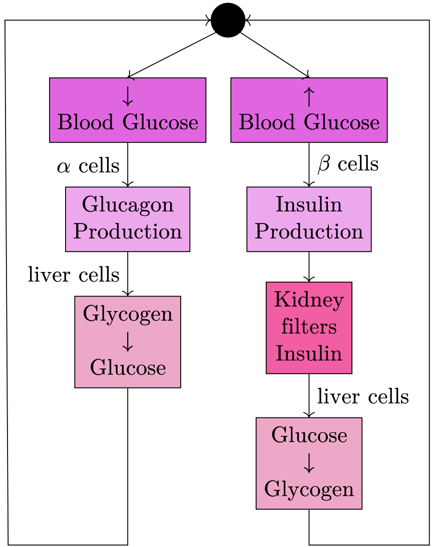
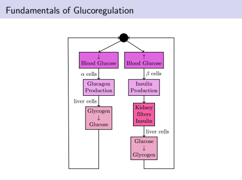
└─Motivation
  └─The Fundamental Problem Addressed
    └─Outline

Outline

1. Motivation
   - The Fundamental Problem Addressed
   - Background
   - Previous Work
2. Results and Contributions
   - Main Results
   - Key Ideas Behind Implementation
3. Summary
   - Takeaways
   - Further Reading

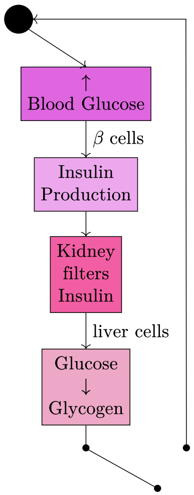
Fundamentals of Glucoregulation





1. The thesis revolves around the functionality of glucoregulation.
2. A high level diagram of the human insulin-glucose system can be described as the image.

Insulin Resistance in ICU Patients



- Stress-induced insulin resistance has been observed in critically ill patients.
- Impaired glycogen production in the liver creates an excess of blood glucose (hyperglycaemia).

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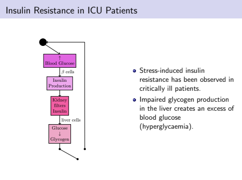

Mention that the break in the chain is just for visual purposes.

Intensive Insulin Therapy

- **What**: Utilizes aggressive insulin infusions and feeding inputs to tightly control blood glucose.
- **Goal**: Prevents hyperglycaemia in patients to reduce mortalities and longer visitation stays.
- **How**: Traditionally has relied on ad-hoc sliding protocols.

Intensive Insulin Therapy

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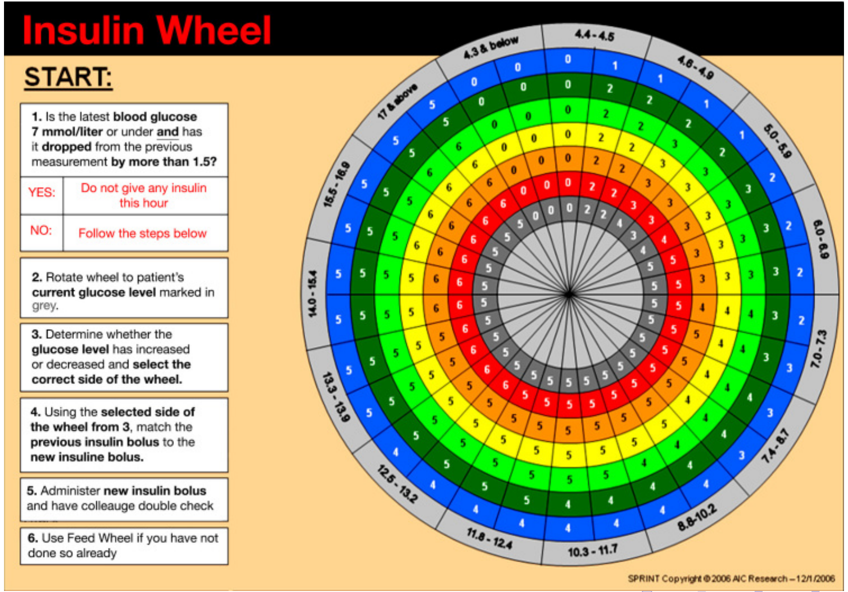
- **How**: Traditionally has relied on ad-hoc sliding protocols.

Overview of Ad-hoc Sliding Protocols



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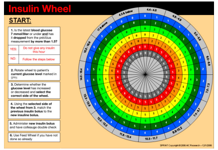
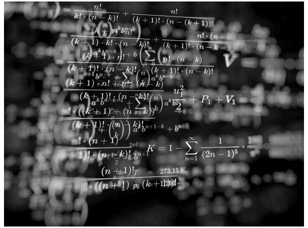
Thesis Presentation
└Motivation
 └The Fundamental Problem Addressed
  └Overview of Ad-hoc Sliding Protocols

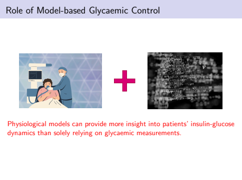


1. Example of the SPRINT Protocol Insulin Wheel (Lonergan et al. 2006).
2. Clinicians administer insulin and feeding from complex flowcharts.
3. Struggles to adapt to the dynamic health of critically ill patients.

Role of Model-based Glycaemic Control



Physiological models can provide more insight into patients' insulin-glucose dynamics than solely relying on glycaemic measurements.



Insulin infusions combined with better understanding of dynamics can provide more insight to the clinicians

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Intensive Control Insulin-Nutrition-Glucose (ICING) model

Lin et al. has developed a clinically-validated nonlinear insulin-glucose model with the following main components

Dynamic	Notation	Value
Blood Glucose	$\dot{G}$	mmol/L
Intersitial Plasma	$\dot{Q}$	mU/L
Plasma Insulin	$\dot{I}$	mU/L
Glucose in the Stomach	$\dot{P}1$	mmol
Glucose in the Gut	$\dot{P}2$	mmol
Endogenous Insulin Production	$u_{en}(t)$	mU/min
Insulin Sensitivity	$S_I(t)$	-
Input	**Notation**	**Value**
Exogenous Insulin	$u_{ex}(t)$	mU/min
Enternal Food	$D(t)$	mmol/min
Parenteral Glucose	$P_N(t)$	mmol/min

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More details on the exact formulation of the dynamics can be read in the paper in further reading.

Computational Approximation of the ICING model

- The ICING model equations are defined in continuous-time
- Discretization of the model is required for the presented estimation methods

For an example dynamic F

$$
\begin{aligned}
t_k &= t_{k-1} + \Delta, \\
\dot{F} &\approx \frac{F(t_k) - F(t_k - \Delta)}{\Delta}, \\
F_k &\approx F_{k-1} + \Delta \dot{F}.
\end{aligned}
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1. According to Euler's method
2. Discretization error is bounded by Δ

Bayesian Filtering

The discrete-time dynamic ICING model can be represented in the stochastic state-space.

State-space Representation

$$x_k \sim p(x_k \mid x_{k-1},\, u_{k-1})$$
$$y_k \sim p(y_k \mid x_k)$$

Parameter Description

System Variable	Notation
Dynamics	x_k
Inputs	u_k
Measurements	y_k

There are various methods of optimal filtering, and each has their own assumptions and limitations

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There are various methods of optimal filtering, and each has their own assumptions and limitations

1. The measurement often depends on the hospital, but will be discussed more later
2. Just the EKF and the PF will be in the topic of this thesis

Approximates nonlinear and Gaussian dynamic and measurement models as a linear transformation.

1. *Prediction step*

$$
\begin{aligned}
m_k^- &= f(m_{k-1},\, u_{k-1}),\\
P_k^- &= F_x(m_{k-1})P_{k-1}F_x^{\mathsf{T}}(m_{k-1}) + F_q(m_{k-1})Q_{k-1}F_q^{\mathsf{T}}(m_{k-1}).
\end{aligned}
$$

2. *Update step*

$$
\begin{aligned}
v_k &= y_k - h(m_k^-),\\
S_k &= H_x(m_k^-)P_k^- H_x^{\mathsf{T}}(m_k^-) + H_r(m_k^-)R_k H_r^{\mathsf{T}}(m_k^-),\\
K_k &= P_k^- H_k^{\mathsf{T}}(m_k^-)S_k^{-1},\\
m_k &= m_k^- + K_k v_k,\\
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Extended Kalman Filter

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Bootstrap Particle Filter

Formulates the posterior of the dynamic function as a Monte Carlo approximation.

1. Sample N particles from the dynamic function

$$x_k^{(i)} \sim p(x_k \mid x_{k-1}^{(i)}, u_{k-1}) \quad i = 1, \ldots, N.$$

2. Update weights from the measurement

$$w_k^{*(i)} \propto p(y_k \mid x_k^{(i)}) \quad i = 1, \ldots, N,$$

$$w_k^{(i)} = \frac{w_k^{*(i)}}{\sum_{j=1}^{N} w_k^{*(j)}} \quad i = 1, \ldots, N.$$

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Parameter Estimation

- There exists unrepresented ICING dynamics $S_{I,k}$ and $u_{en,k}$.
- Naively assigning either as a general population constant can hinder the performance of the estimators.

State Augmentation

1. Add artificial process noise to the parameters of interest θ_k

$$\theta_k = \theta_{k-1} + \epsilon_{k-1}, \qquad \text{with } \epsilon_{k-1} \sim \mathcal{N}(0,\, W_{k-1}).$$

2. Augment the parameters to the existing dynamics

$$\tilde{x}_k = [x_k, \theta_k]^\top .$$

3. Update the state-space representation

$$\begin{aligned} \tilde{x}_k &\sim p(\tilde{x}_k \mid \tilde{x}_{k-1},\, u_{k-1}), \\ y_k &\sim p(y_k \mid \tilde{x}_k). \end{aligned}$$

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Formulation of Synthetic Patients

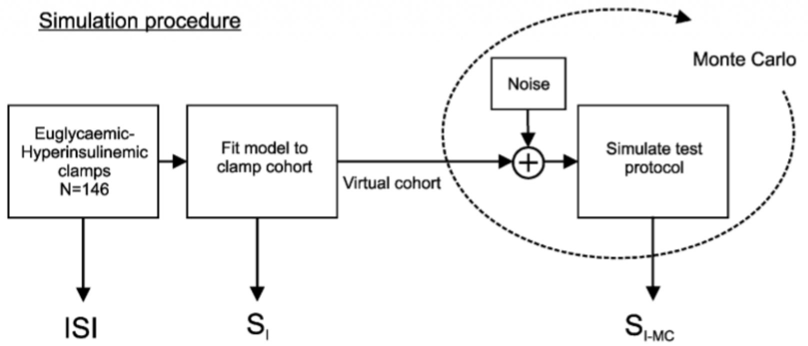

Figure: An example of a simulation procedure (Lotz et al. 2008).

- Synthetic patient trajectories model clinical insulin–glucose dynamics.
- Existing methods typically perturb previously recorded clinical data.

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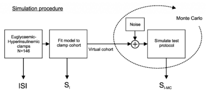

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Estimation of Patient Insulin-Glucose Dynamics

- Previous studies have employed Bayesian filtering methods:
  - Extended and unscented Kalman filters
  - Particle filters
- Practical implementation is often insufficiently addressed:
  - Constant nutrition and insulin inputs
  - Known initial insulin–glucose dynamics
- Estimator configurations and performance are not systematically compared.

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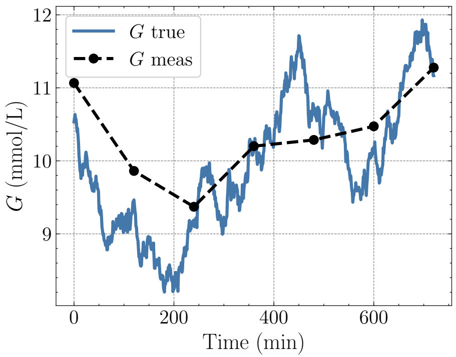
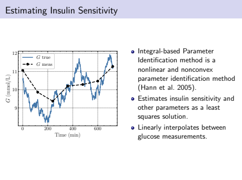
Estimating Insulin Sensitivity



- Integral-based Parameter Identification method is a nonlinear and nonconvex parameter identification method (Hann et al. 2005).
- Estimates insulin sensitivity and other parameters as a least squares solution.
- Linearly interpolates between glucose measurements.



1. Doesn't account for noise in measurements.

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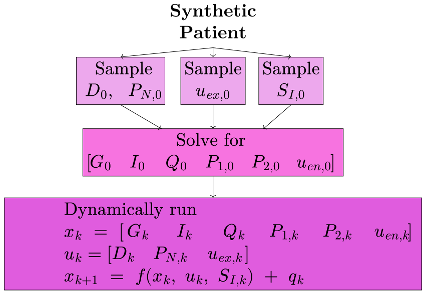
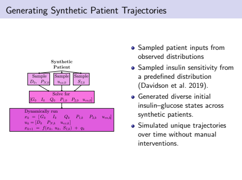
Generating Synthetic Patient Trajectories



- Sampled patient inputs from observed distributions
- Sampled insulin sensitivity from a predefined distribution (Davidson et al. 2019).
- Generated diverse initial insulin–glucose states across synthetic patients.
- Simulated unique trajectories over time without manual interventions.



Synthetic Patient Variability

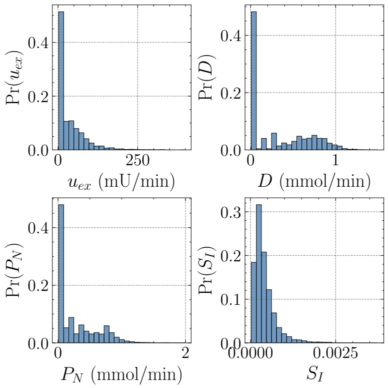


Figure: Initial inputs and insulin sensitivity.

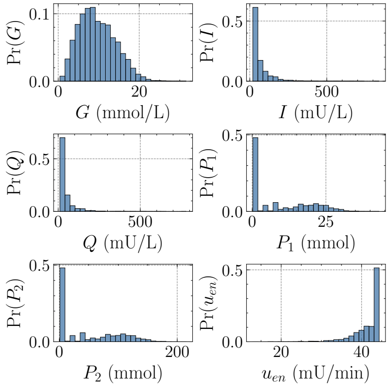


Figure: Initial insulin-glucose states.

- The empirical histograms reflect variability in baseline insulin-glucose dynamics.

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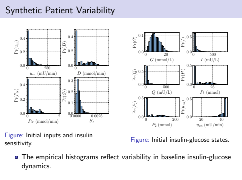
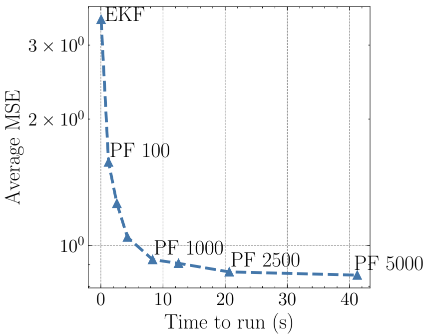
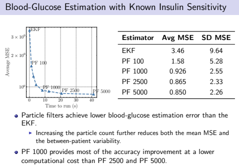


Blood-Glucose Estimation with Known Insulin Sensitivity



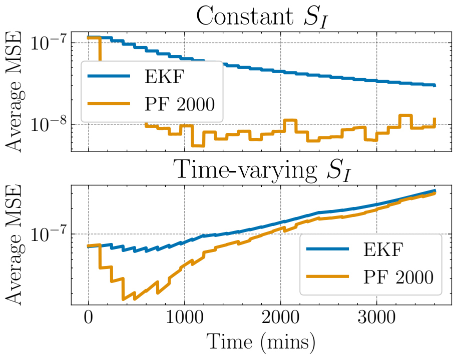
Estimator	Avg MSE	SD MSE
EKF	3.46	9.64
PF 100	1.58	5.28
PF 1000	0.926	2.55
PF 2500	0.865	2.33
PF 5000	0.850	2.26

- Particle filters achieve lower blood-glucose estimation error than the EKF.
    - Increasing the particle count further reduces both the mean MSE and the between-patient variability.
- PF 1000 provides most of the accuracy improvement at a lower computational cost than PF 2500 and PF 5000.



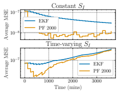
1. MSE shown on a logarithmic scale.
2. MSE and SD is averaged over patients.
3. PF lowers both the mean error and the spread.
4. These results assume we know the insulin sensitivity.

Estimating Constant and Time-Varying Insulin Sensitivity



- For constant S_I, PF 2000 reduces the estimation MSE substantially faster than the EKF and achieves a lower steady-state error.
- For time-varying S_I, both methods show increasing MSE, indicating difficulty tracking changes in insulin sensitivity.

Estimating Constant and Time-Varying Insulin Sensitivity



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Design of Synthetic Patient Trajectories

- Real ICU insulin and feeding records from Karolinska University Hospital are used to generate realistic synthetic patient trajectories:
  1. 27,113 ICU patients
  2. 39.5% of patients received insulin during their ICU stay
  3. Approximately 100 million minutes of recorded feeding data
  4. Approximately 80 million minutes of recorded insulin data
- Parenteral glucose and insulin inputs follow periodic exponential ramps.
- The feeding input is held constant.
- Time-varying insulin sensitivity is represented by a piecewise-linear function.

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ICING Model Configuration

- The ICING model is discretized with a time step of $\Delta t = 1$ minute.
- Blood-glucose measurements are available every 120 minutes:

$$f_{\mathrm{BG}} = \frac{1}{120}\ \mathrm{min}^{-1}.$$

Measurement noise is additive Gaussian:

$$r_k \sim \mathcal{N}(0, 0.25^2).$$

- Process noise is assigned as 10% of the corresponding initial-state variances:

$$q_k \sim \mathcal{N}\left(\mathbf{0}, \mathrm{diag}(\boldsymbol{\sigma}^2)\right).$$

- Insulin sensitivity is modeled as a positive, piecewise-varying parameter over 1200-minute intervals:

$$\log S_{I,k+1} = \log S_{I,k} + q_{S_I,k}, \qquad q_{S_I,k} \sim \mathcal{N}\left(0, \frac{\sigma_{S_I}^2}{1200}\right), \quad \sigma_{S_I} = \frac{1}{3}.$$

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State Estimation Evaluation Procedure

- The evaluation uses $N = 250$ unique synthetic patients.
- The simulation lasted 12 hours to allow the estimators to converge.
- EKF and PFs containing (100, 250, 500, 1000, 1500, 3000, 5000) particles were used to compare accuracy and the average time to run the simulation.
- MSE is averaged across patients according to each blood-glucose estimate $\hat{G}$ of length L:

$$\text{Avg MSE} = \frac{1}{NL}\sum_{i=1}^{N}\sum_{j=1}^{L}\left(G_i[j] - \hat{G}_i[j]\right)^2,$$

$$\text{SD MSE} = \frac{1}{N-1}\sum_{i=1}^{N}\left(\frac{1}{L}\sum_{j=1}^{L}\left(G_i[j] - \hat{G}_i[j]\right)^2 - \text{Avg MSE}\right)^2$$

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Parameter-Estimation Evaluation Procedure

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- Each patient is simulated for 60 hours to capture the relatively slow changes in insulin sensitivity.
- State-augmented EKF and PF 2000 estimators are compared under two scenarios:
  - constant insulin sensitivity
  - time-varying insulin sensitivity
- The insulin-sensitivity MSE is computed at each time step:

$$\mathrm{MSE}_{S_{i,k}} = \frac{1}{N}\sum_{i=1}^{N}\left(S_{I,i}[k]-\hat{S}_{I,i}[k]\right)^2.$$

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Outline

2026-09-08

Thesis Presentation
└Summary
 └Takeaways
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1. Motivation
   - The Fundamental Problem Addressed
   - Background
   - Previous Work
2. Results and Contributions
   - Main Results
   - Key Ideas Behind Implementation
3. Summary
   - Takeaways
   - Further Reading

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References

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